

Lab Assignment 04



Inspiring Excellence

Course Code:	CSE111
Course Title:	Programming Language II
Topic:	Reference Passing & Multiclass Problem
Number of Tasks:	11 (Classwork: 05, Homework: 06)

[Submit all the Coding Tasks (Homework: Task 1 to 4) in the Google Form shared on buX before the next lab. Submit the Tracing Tasks (Homework: Task 5 to 6) handwritten to your Lab Instructors at the beginning of the lab]

[You are not allowed to change the driver codes of any of the tasks]

CLASSWORK

Task 1

Design the **Triangle** Class that will produce the following output. We will consider both triangles to have the same sides if all sides are equal in the same orientation/sequence only.

Types of Triangle:

- Equilateral: When all sides in the same orientation are equal.
- Isosceles: When any two sides of a triangle in the same orientation are equal.
- Scalene: When all sides are of different lengths.

Driver Code	Output
<pre>public class TriangleTester{ public static void main(String args[]){ Triangle t1 = new Triangle(4, 4, 4); Triangle t2 = new Triangle(4, 5, 6); Triangle t3 = new Triangle(4, 5, 6); Triangle t4 = new Triangle(5, 4, 6); t1.triangleDetails(); System.out.println("-----1-----"); System.out.println(t1.printTriangleType()); System.out.println("-----2-----"); t3.triangleDetails(); System.out.println(t3.printTriangleType()); System.out.println("-----3-----"); t4.triangleDetails(); System.out.println(t4.printTriangleType()); System.out.println("-----4-----"); t2.compareTriangles(t3); System.out.println("-----5-----"); t1.compareTriangles(t2); System.out.println("-----6-----"); t1 = t2; t1.compareTriangles(t2); System.out.println("-----7-----"); t3.compareTriangles(t4); } }</pre>	<pre>Three sides of the triangle are: 4, 4, 4 Perimeter: 12 -----1----- This is an Equilateral Triangle. -----2----- Three sides of the triangle are: 4, 5, 6 Perimeter: 15 This is a Scalene Triangle. -----3----- Three sides of the triangle are: 5, 4, 6 Perimeter: 15 This is a Scalene Triangle. -----4----- Addresses are different but the sides of the triangles are equal. -----5----- Addresses, length of the sides and perimeter all are different. -----6----- These two triangle objects have the same address. -----7----- Only the perimeter of both triangles is equal.</pre>

Task 2

Design the **Shop** class with necessary properties so that the following output is produced.

Driver Code	Output
<pre>public class ShopTester{ public static void main(String[] args){ Shop s1 = new Shop("Agora", 4); Shop s2 = new Shop(3); System.out.println("1====="); s1.info(); System.out.println("2====="); s2.info(); System.out.println("3====="); s1.addItem("KitKat", 25); s1.addItem("Hershey's", 100); s1.addItem("Dairy Milk", 150); s1.addItem("Feastables", 500); s1.addItem("Prime", 1200); System.out.println("4====="); s1.info(); System.out.println("5====="); s1.purchase("KitKat"); s1.purchase("Snickers"); System.out.println("6====="); s2.info(); System.out.println("7====="); s2.addItem(s1, "KitKat"); s2.addItem("Toblerone", 80); s2.addItem(s1, "Mars"); System.out.println("8====="); s2.info(); } }</pre>	<pre>Agora shop created! Mega shop created! 1===== Shop Name: Agora Item Details: No items in shop 2===== Shop Name: Mega Item Details: No items in shop 3===== Prime, could not be added 4===== Shop Name: Agora Item Details: 4 / 4 KitKat - 25 Tk Hershey's - 100 Tk Dairy Milk - 150 Tk Feastables - 500 Tk 5===== Purchase Complete! Snickers, is not available in this shop 6===== Shop Name: Mega Item Details: No items in shop 7===== Mars, not found in Agora! 8===== Shop Name: Mega Item Details: 2 / 3 KitKat - 25 Tk Toblerone - 80 Tk</pre>

Task 3

Write the **Teacher** and **Course** classes so that the TestTeacher class produces the outputs given.

Hint: A teacher can add a maximum of 3 courses.

Driver Code	Output
<pre>public class TestTeacher{ public static void main(String [] args){ Teacher t1 = new Teacher("Matin Saad Abdullah","MSA"); Teacher t2 = new Teacher("Mumit Khan","MMK"); Teacher t3 = new Teacher("Sadiah Hamid Kazi","SKZ"); Course c1 = new Course("CSE 110"); Course c2 = new Course("CSE 111"); Course c3 = new Course("CSE 220"); Course c4 = new Course("CSE 221"); Course c5 = new Course("CSE 230"); Course c6 = new Course("CSE 310"); Course c7 = new Course("CSE 320"); Course c8 = new Course("CSE 340"); t1.addCourse(c1); t1.addCourse(c2); t2.addCourse(c3); t2.addCourse(c4); t2.addCourse(c5); t3.addCourse(c6); t3.addCourse(c7); t3.addCourse(c8); System.out.println("1====="); t1.printDetail(); System.out.println("2====="); t2.printDetail(); System.out.println("3====="); t3.printDetail(); } }</pre>	<pre>A new teacher has been created A new teacher has been created A new teacher has been created 1===== Name: Matin Saad Abdullah Initial: MSA List of courses: CSE 110 CSE 111 2===== Name: Mumit Khan Initial: MMK List of courses: CSE 220 CSE 221 CSE 230 3===== Name: Sadiah Hamid Kazi Initial: SKZ List of courses: CSE 310 CSE 320 CSE 340</pre>

Task 4

Design the **Team** and **Player** classes to get the output as shown.

Hints:

- Create an array in the Team class to store the Player's object
- Use constructor overloading technique for Team class
- A team can add maximum 11 players

Driver Code	Output
<pre>public class TeamTester { public static void main(String[] args) { Team b = new Team(); b.updateName("Bangladesh"); Player mashrafi = new Player("Mashrafi", 42, 100); b.addPlayer(mashrafi); Player tamim = new Player("Tamim", 35, 70); b.addPlayer(tamim); System.out.println("====="); b.printDetail(); System.out.println("====="); Team a = new Team("Australia"); Player ponting = new Player("Ponting", 50, 300); a.addPlayer(ponting); Player lee = new Player("Lee", 49, 200); a.addPlayer(lee); a.printDetail(); } }</pre>	<pre>===== Team: Bangladesh List of players: Name: Mashrafi Age: 42, Total Matches: 100 Name: Tamim Age: 35, Total Matches: 70 ===== Team: Australia List of players: Name: Ponting Age: 50, Total Matches: 300 Name: Lee Age: 49, Total Matches: 200</pre>

Task 5

1	public class Trace{
2	public int x, y = 5, temp = -4, sum = 3;
3	public Trace(){
4	y = temp + 3 ;
5	sum = 3 + temp + 2;
6	temp -= 2;
7	}
8	public Trace(Trace t){
9	sum = t.sum++;
10	x = t.x + 2;
11	t.methodB(2,3);
12	}
13	public void methodA(int m, int n){
14	int x = 2;
15	y = y + m + (temp++);
16	x = x + 5 + n;
17	sum = sum + x + y;
18	System.out.println(x + " " + y+ " " + sum);
19	}
20	public void methodB(int m, int n){
21	int y = 0;
22	y = y + this.y;
23	x = this.y + 2 + temp;
24	methodA(x, y);
25	sum = x + y + sum;
26	System.out.println(x + " " + y+ " " + sum);
27	}
28	}

<pre> public class Tester5 { public static void main(String args []){ Trace t1 = new Trace(); Trace t2 = new Trace(t1); t1.methodA(1, 2); t2.methodB(3, 2); } } </pre>	Outputs		

HOMEWORK

Task 1

Design the **Player** class to obtain the following output from the driver code. You can assume, each player can defeat at most 5 villains.

Driver Code	Expected Output
<pre>public class PlayerTester { public static void main(String[] args) { Player ben = new Player("Ben", 10); System.out.println("=====1====="); ben.viewInfo(); System.out.println("=====2====="); ben.defeatVillain("Vilgax", 100); System.out.println("=====3====="); ben.defeatVillain("Yamcha", 10); System.out.println("=====4====="); ben.viewInfo(); System.out.println("=====5====="); ben.defeatVillain('8', "Vilgax", 100); System.out.println("=====6====="); ben.viewInfo(); System.out.println("=====7====="); Player kevin = new Player("Kevin"); System.out.println("=====8====="); kevin.viewInfo(); System.out.println("=====9====="); ben.defeatVillain(kevin); System.out.println("=====10====="); ben.viewInfo(); System.out.println("=====11====="); Player goku = new Player("Goku", 9000); System.out.println("=====12====="); ben.defeatVillain(goku); } }</pre>	<pre>Ben joined the game HP: 10 =====1===== Player Name: Ben Current HP: 10 =====2===== failed to defeat Vilgax =====3===== defeated Yamcha =====4===== Player Name: Ben Current HP: 20 Defeated: Yamcha, =====5===== HP with 8x boost: 160 defeated Vilgax =====6===== Player Name: Ben Current HP: 260 Defeated: Yamcha, Vilgax, =====7===== Kevin joined the game HP: 100 =====8===== Player Name: Kevin Current HP: 100 =====9===== defeated Kevin =====10===== Player Name: Ben Current HP: 360 Defeated: Yamcha, Vilgax, Kevin, =====11===== Goku joined the game HP: 9000 =====12===== failed to defeat Goku</pre>

Task 2

Write “**Student**” class to show the following expected outputs

Note:

- ❖ A student can't take any course until the CGPA is set.
- ❖ A student cannot take more than 4 courses.
- ❖ A student with CGPA below 3 cannot take more than 3 courses.

Driver Code	Expected Output
<pre>public class StudentDriver { public static void main(String[] args){ Student student1 = new Student(12345678); System.out.println("1-----"); student1.addCourse("CSE110"); System.out.println("2-----"); student1.storeCG(2.5); student1.addCourse("CSE110"); student1.addCourse("ENG101"); student1.showAdvisee(); System.out.println("3-----"); student1.removeAllCourse(); student1.showAdvisee(); System.out.println("4-----"); student1.storeID(54652365); student1.addCourse(new String [] {"SOC101","CSE111","ENG102"}); student1.showAdvisee(); System.out.println("5-----"); student1.addCourse("CSE230"); student1.showAdvisee(); System.out.println("6-----"); Student student2 = new Student(975738383,3.7); System.out.println("7-----"); String[] courses2 = {"CSE220","PHY112","MAT120","BUS101","CHN101"}; student2.addCourse(courses2); student2.showAdvisee(); } }</pre>	<pre>A student with ID 12345678 has been created. 1----- Failed to add CSE110 Set CG first 2----- Student ID: 12345678, CGPA: 2.5 Added courses are: CSE110 ENG101 3----- Student ID: 12345678, CGPA: 2.5 No courses added. 4----- Student ID: 54652365, CGPA: 2.5 Added courses are: SOC101 CSE111 ENG102 5----- Failed to add CSE230 CG is low. Can't add more than 3 courses. Student ID: 54652365, CGPA: 2.5 Added courses are: SOC101 CSE111 ENG102 6----- A student with ID 975738383 and cgpa 3.7 has been created. 7----- Failed to add CHN101 Maximum 4 courses allowed. Student ID: 975738383, CGPA: 3.7 Added courses are: CSE220 PHY112 MAT120 BUS101</pre>

Task 3

Design the **TravelPrep** class so that the given output is generated.

- This class keeps track of the package info and the destination that goes into it.
- The initial budget will be 1250 yen for any traveller.
- A traveller can visit at most 3 destinations if that falls under their budget.

Driver Code	Expected Output
<pre> public class TravelPlan { public static void main(String[] args) { TravelPrep t1 = new TravelPrep(); System.out.println("1====="); TravelPrep t2 = new TravelPrep("Package-2", 2100); System.out.println("2====="); TravelPrep d1 = new TravelPrep("Fushimi", "Shrine"); TravelPrep d2 = new TravelPrep("Lake", "Kawaguchi", 550); TravelPrep d3 = new TravelPrep("Shrine", "Hieizan", 1000); TravelPrep d4 = new TravelPrep("Lake", "Ashi", 620); System.out.println("3====="); t2.add_to_itinerary(d1); t2.add_to_itinerary(d2, d3); t2.show_itinerary(); System.out.println("4====="); System.out.println(d2.updateCost(60)); System.out.println("5====="); t1.t_name = "Package-1"; t1.add_to_itinerary(d2, d4); t1.add_to_itinerary(d3); System.out.println("6====="); t1.show_itinerary(); } } </pre>	<pre> Unknown package would cost at most 1250 yen 1===== Package-2 would cost at most 2100 yen 2===== Fushimi Shrine costs 300 yen Kawaguchi Lake costs 550 yen Hieizan Shrine costs 1000 yen Ashi Lake costs 620 yen 3===== Itinerary for Package-2 1. Fushimi Shrine - 300 yen 2. Kawaguchi Lake - 550 yen 3. Hieizan Shrine - 1000 yen Total cost of 3 destinations: 1850 yen 4===== Cost of Kawaguchi Lake updated to 610 5===== Budget going overboard 6===== Itinerary for Package-1 1. Kawaguchi Lake - 610 yen 2. Ashi Lake - 620 yen Total cost of 2 destinations: 1230 yen </pre>

Task 4

Design the **Student** and the **Connect** class so that the following output is produced.

Note:

- A student's email, password, and login status are null by default while creating an object of the Student class.
- Your code should satisfy the conditions mentioned in the output only.
- Connect class will have two instance variables: totalAdvisee and an array of Student type to store the student object. The array will be updated inside the advising() method only when the advising is successful.
- Connect can take at most 5 advisees.

Driver Code	Expected Output
<pre>public class ConnectTester { public static void main(String[] args) { Student rakib = new Student("Rakib", 12301455, "CSE"); Student roy = new Student("Roy", 12501345, "CS"); System.out.println("1*****"); Connect connectObj = new Connect(); System.out.println("2*****"); connectObj.login(rakib); System.out.println("3*****"); connectObj.advising(rakib); System.out.println("4*****"); rakib.email = "rakib@hotmail.com"; rakib.password = "1234"; System.out.println("5*****"); connectObj.login(rakib); System.out.println("6*****"); connectObj.advising(rakib); System.out.println("7*****"); connectObj.advising(rakib, "CSE110", "PHY111", "MAT110", "CSE260"); System.out.println("8*****"); connectObj.advising(rakib, "CSE110", "PHY111", "MAT110"); System.out.println("9*****"); connectObj.allAdviseeInfo(); System.out.println("10*****"); roy.email = "roy@hotmail.com"; roy.password = "abcd"; connectObj.login(roy); System.out.println("11*****"); connectObj.advising(roy, "CSE110", "ENG101", "PHY112"); System.out.println("12*****"); connectObj.allAdviseeInfo(); } }</pre>	<pre>Student object is created Student object is created 1***** Connect is ready to use! 2***** Email and password need to be set. 3***** Please login to advise courses! 4***** 5***** Login successful 6***** You haven't selected any courses. 7***** You need special approval to take more than 3 courses. 8***** Advising successful! 9***** Total Advisee: 1 Name: Rakib ID: 12301455 Department: CSE Advised Courses: CSE110 PHY111 MAT110 ===== 10***** Login successful 11***** Advising successful! 12***** Total Advisee: 2 Name: Rakib ID: 12301455 Department: CSE Advised Courses: CSE110 PHY111 MAT110 ===== Name: Roy ID: 12501345 Department: CS Advised Courses: CSE110 ENG101 PHY112 =====</pre>

Task 5

1	public class QuizA {
2	public int x, y;
3	public int sum = 1;
4	public QuizA(int x, int y) {
5	this.x = y;
6	this.y = x;
7	}
8	public void methodA() {
9	int x = 3;
10	y = this.y + x;
11	QuizA exam = new QuizA(5, 11);
12	exam.sum = x;
13	exam.y = this.y;
14	x = this.x + x + exam.sum;
15	this.y = this.sum + methodB(exam.sum, exam);
16	System.out.println(exam.x + " " + this.y + " " + sum);
17	sum = x % 2 + this.x;
18	y = x + y + exam.sum;
19	System.out.println(x + " " + y + " " + sum);
20	}
21	public int methodB(int x1, QuizA x2) {
22	int y = 0;
23	y = this.y + x2.sum;
24	x2.sum = x1 + x2.x;
25	sum = sum + x + y;
26	System.out.println(this.x + " " + this.y + " " + sum);
27	return x2.sum;
28	}
29	}

Driver Code	Output		
<pre> public class QuizTesterA{ public static void main(String []largs){ QuizA q1 = new QuizA(3,4); q1.methodA(); } } </pre>			

Task 6

1	public class A {
2	public int x, y = 2, sum = 5;
3	public A(){
4	x = sum - y;
5	this.methodB(1);
6	}
7	public A(int x){
8	this.x = x;
9	}
10	public void methodB(int y) {
11	int temp = (y++) + (++this.y);
12	sum = y + temp - ++this.x;
13	System.out.println(y + " " + temp + " " + (sum++));
14	this.y = (++temp) + methodC(temp, y) + x;
15	}
16	public void methodA(int temp, int x) {
17	A a2 = new A(6);
18	a2.sum = this.methodC(a2);
19	System.out.println(this.x + " " + a2.sum + " " + y);
20	this.sum = a2.methodC(this);
21	System.out.println(a2.x + " " + this.sum + " " + a2.y);
22	}
23	public int methodC(A a) {
24	return this.x + a.y + a.sum;
25	}
26	public int methodC(int sum, int y) {
27	y = (this.x++) + sum + 3;
28	return y;
29	}
30	}

<pre> public class TesterA{ public static void main(String [] args) { A a1 = new A(); a1.methodA(4, 3); } } </pre>	Outputs		

Ungraded Tasks (Optional)

(You don't have to submit the ungraded tasks)

Task 1

Driver Code	Output
<pre>public class TaxiTester{ public static void main(String args []){ Taxi t1 = new Taxi("MX8190", "Gulshan"); Taxi t2 = new Taxi("MX6718", "Badda"); Taxi t3 = new Taxi("MX2789", "Gulshan"); System.out.println("1-----"); t1.compareTaxi(t2); System.out.println("2-----"); t1.compareTaxi(t3); } }</pre>	<pre>1----- Routes are not the same! 2----- Both routes are the same!</pre>

Task 2

Driver Code	Output
<pre>public class ConnectTester{ public static void main(String[] args) { ConnectFriends sanaf = new ConnectFriends("Sanaf"); System.out.println("=====1====="); ConnectFriends mantasha = new ConnectFriends("Mantasha", 3); ConnectFriends mostafiz = new ConnectFriends("Mostafiz"); ConnectFriends matt = new ConnectFriends("Matt", 4); System.out.println("=====2====="); sanaf.sendFriendRequest(mantasha); System.out.println("=====3====="); sanaf.sendFriendRequest(mostafiz, matt); System.out.println("=====4====="); sanaf.showDetails(); System.out.println("=====5====="); sanaf.removeRequest("Mantasha"); System.out.println("=====6====="); sanaf.showDetails(); System.out.println("=====7====="); sanaf.removeRequest("Murdock"); System.out.println("=====8====="); sanaf.removeRequest("Matt"); sanaf.removeRequest("Mostafiz"); sanaf.showDetails(); System.out.println("=====9====="); mantasha.showDetails(); } }</pre>	<pre>Welcome to ConnectFriends, Sanaf =====1===== Welcome to ConnectFriends, Mantasha Welcome to ConnectFriends, Mostafiz Welcome to ConnectFriends, Matt =====2===== Sanaf sent a friend request to Mantasha. =====3===== Sanaf sent a friend request to Mostafiz. Sanaf has reached the friend request limit! =====4===== User Name: Sanaf Maximum number of Sent Friend Request: 2 Total Friends Request: 2 Sent Friends Request: Mantasha Mostafiz =====5===== Reuquest to add Mantasha is removed for Sanaf. =====6===== User Name: Sanaf Maximum number of Sent Friend Request: 2 Total Friends Request: 1</pre>

}	Sent Friends Request: Mostafiz =====7===== Murdock is not in Sanaf's sent request list. =====8===== Matt is not in Sanaf's sent request list. Reuquest to add Mostafiz is removed for Sanaf. User Name: Sanaf Maximum number of Sent Friend Request: 2 Total Friends Request: 0 Sent Friends Request: =====9===== User Name: Mantasha Maximum number of Sent Friend Request: 3 Total Friends Request: 0 Sent Friends Request:
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Task 3

Driver Code	Output
<pre> public class Tester{ public static void main(String args[]){ Cart c1 = new Cart(); Cart c2 = new Cart(); Cart c3 = new Cart(); Cart cartHub = new Cart(); c1.storeProduct("Pen", 10); c2.storeProduct("Pencil", 15, "Book", 100); c2.storeProduct("Eraser", 5); c3.storeProduct("Sharpener", 8); System.out.println("1-----"); c1.info(); System.out.println("2-----"); c2.info(); System.out.println("3-----"); c1.transferItem(c2, "Pen"); System.out.println("4-----"); c1.info(); System.out.println("5-----"); c2.info(); System.out.println("6-----"); cartHub.addCart(c1,c2,c3); System.out.println("7-----"); cartHub.allCartInfo(); } } </pre>	<pre> 1----- Cart Items: Pen - 10 2----- Cart Items: Pencil - 15 Book - 100 Eraser - 5 3----- Item transferred successfully. 4----- Cart Items: The cart is empty. 5----- Cart Items: Pencil - 15 Book - 100 Eraser - 5 Pen - 10 6----- 7----- Cart Items: The cart is empty. Cart Items: Pencil - 15 </pre>

<pre> } } </pre>	Book - 100 Eraser - 5 Pen - 10 Cart Items: Sharpener - 8
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Task 4

Driver Code	Output
<pre> public class TaxiTester{ public static void main(String args []){ Taxi t1 = new Taxi("MX8190", "Gulshan"); Passenger p1 = new Passenger("Ali", 20); Passenger p2 = new Passenger("Mifta", 45); Passenger p3 = new Passenger("Marzan", 32); System.out.println("1-----"); t1.showTaxi(); System.out.println("2-----"); p1.info(); p2.info(); System.out.println("3-----"); t1.addPassenger(p1); t1.addPassenger(p2); t1.addPassenger(p3); System.out.println("4-----"); t1.showTaxi(); } } </pre>	<pre> 1----- Taxi No: MX8190 Route: Gulshan Passenger List: 2----- Name: Ali Age: 20 Name: Mifta Age: 45 3----- 4----- Taxi No: MX8190 Route: Gulshan Passenger List: Ali 20 Mifta 45 Marzan 32 </pre>

Task 5

1	public class TracingX {
2	public int x, y = 1;
3	public int metA(int y){
4	y += x + 3;
5	int temp = y + this.y;
6	if (temp % 2 == 0){
7	return temp;
8	}
9	TracingX t = new TracingX();
10	t.y = this.x - (++x) + t.x;
11	this.y = y + t.metA(t.x);
12	System.out.println(x + " " + y + " " + temp);
13	return temp + this.y;
14	}
15	}

Driver code:

```
public class TesterX {  
    public static void main(String[] args) {  
        TracingX t1 = new TracingX();  
        t1.y = t1.x = 5;  
        TracingX t2 = new TracingX();  
        t2.x = t1.metA(2);  
        t2.y = t2.metA(4);  
        System.out.println(t1.y + t1.x + " " + t2.x + " " + t2.y);  
    }  
}
```

Output:

Task 6

1	public class TestX{
2	public int result = 17, i, n = 7, m = 5;
3	public int [] num = {6,8,11};
4	public void compute(int [] nums){
5	n = nums[0] + this.m;
6	this.result += nums[i];
7	TestX temp = new TestX();
8	boolean check = !(temp.modify(temp, num));
9	if (check) {
10	m = nums[1] + temp.result;
11	} else {
12	m = result - nums[3];
13	}
14	System.out.println(m + " " + n + " " + result);
15	}
16	public boolean modify(TestX obj, int[] arr){
17	arr[1]++;
18	i=i+1;
19	obj.m = this.result + arr[i];
20	obj.result = obj.m - obj.n + n;
21	System.out.println(obj.m + " " + obj.n + " " + result);
22	return arr[1] % 2 == 0;
23	}
24	public boolean modify(TestX obj){
25	num[i]++;
26	i=i+1;
27	obj.m = this.result + num[i];
28	obj.result = num[2] - obj.n + n;
29	System.out.println(obj.m + " " + obj.n + " " + result);
30	return num[1] % 2 != 0;
31	}
32	}

<pre> public class tester12{ public static void main(String[] args){ TestX tx = new TestX(); int[] data = {4, 6, 2, 3, 5}; tx.compute(data); tx.modify(tx); } } </pre>	Outputs		